

TOPICAL REVIEW

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Caring for Patients With Diabetes in Stroke Neurology

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ABSTRACT: Diabetes is a heterogeneous disease that affects 9% of the world's population (11% in the United States). The consequences of diabetes for the brain are severe; it nearly doubles a person's risk of stroke and is a major contributor to risk for cerebral small vessel disease and dementia. These effects on the brain are in addition to peripheral neuropathy, retinopathy, nephropathy, and coronary heart disease. In this article, we explain the treatments that can prevent or mitigate its harmful effects and propose a role for neurologists and other neurology clinicians in managing patients during routine care.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: brain ■ cognitive dysfunction ■ cohort studies ■ dementia ■ heart disease

Diabetes is a common and under-treated factor in brain health. Cohort studies first quantified the risk associated with diabetes for stroke and heart disease in the 1970s and for cognitive dysfunction in the 1990s. Despite growing concern among health leaders, the prevalence and harms caused by diabetes burgeoned from the 1970s onward as a result of social and economic changes leading to increased caloric intake and reduced physical activity.^{1,2} Little could be done to thwart the social and economic forces behind diabetes, but in 1993, scientists first identified therapies to mitigate its harm to blood vessels.³ In that year, scientists discovered a threshold for glucose control in type 1 diabetes (T1D) associated with prevention of microvascular complications in eyes, kidneys, and nerves. Evidence in type 2 diabetes (T2D) followed shortly.⁴ Other trials showed the benefit of cholesterol and blood pressure reduction for preventing macrovascular complications, including stroke and myocardial infarction.^{5–8} In 2015 to 2016, a major milestone was passed with the recognition that glucose-lowering drugs from 2 new classes safely reduced risk for cardiovascular disease in patients with T2D, SGLT2 (sodium glucose co-transporter 2) inhibitors, and GLP-1 (glucagon-like peptide 1) receptor agonists.

The challenge for those who care for patients with diabetes and stroke or high risk for stroke is to assure that these advances reach their patients.

EPIDEMIOLOGY

Diabetes in the General Population

There are 2 main types of diabetes with distinct causes and treatments.⁹ T1D is primarily an autoimmune disease leading to the destruction of pancreatic islet cells, resulting in loss of insulin production and secretion. Most patients are diagnosed as children or young adults. They require life-long insulin replacement therapy to avoid hyperglycemia and ketoacidosis. T2D is usually diagnosed later in life and is caused by a progression of resistance to insulin action in peripheral tissues and a relative reduction in insulin output by the pancreas. At least at its early stages, T2D is potentially reversible with exercise, diet, and weight reduction to improve insulin resistance. Primary care clinicians can manage most patients with T2D. In the United States, 5% to 10% of patients with diabetes have T1D, and 90% to 95% have T2D.^{9,10}

Diabetes of either type affects 9% to 15% of US adults aged 18 years and older.¹¹ The prevalence of T1D does not vary by age (ie, it remains about 0.5% after age 18 years), but the prevalence of T2D rises markedly, reaching 20% to 27% for persons in the United States over 65 years.¹⁰ In the United States, the prevalence of T2D is higher for non-Hispanic Black (15%) and

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Hispanic individuals (15%), compared with non-Hispanic Asian (12%) and White individuals (8%).^{10,11}

There is some geographic variation in incidence of childhood T1D, but the main risk factors are race, ethnicity (higher in non-Hispanic whites) and family history. Other than age, the most important independent risk factor for T2D is obesity.¹² Other independent risk factors for T2D (ie, family history, low levels of physical activity, poor quality diet, race, ethnicity, and socioeconomic factors) interact with obesity in complex ways to modify risk. Because of the importance of obesity, inactivity and diet, most persons could potentially reduce their risk for T2D by a healthy lifestyle.¹² In fact, patients who lose enough weight can achieve disease remission and discontinue glucose-lowering drugs.¹³

Diabetes of either type is one of the most prevalent risk factors for both coronary heart disease and stroke. In a recent meta-analysis, the adjusted hazard ratio for coronary heart disease was 1.84 (95% CI, 1.66–2.04) in patients with diabetes (T1D or T2D) compared with patients without diabetes.¹⁴ Estimates of the hazard ratio (or relative risk) for stroke are similar, ranging from 1.5 to 2.^{15–17} Sex and race modify the effect of diabetes on risk for stroke.^{15,16,18–21} The relative risk for stroke is higher for women than men^{15,22} and higher for Black persons than White persons.^{18,21} Diabetes is responsible for at least 10% (75 000) of strokes per year in the United States.²³

Two other aspects of diabetes in the general population warrant mention. First, because prevalence is higher among Hispanic and Black persons in the United States,¹⁰ it is one cause for disparity in rates of stroke and heart disease.^{24,25} Second, there is evidence that the effect of diabetes on risk for cardiovascular and cerebrovascular disease is falling, possibly because of improved management of certain risk factors including hypertension, cholesterol, and smoking.^{2,26} The US Centers for Disease Control and Prevention reported a decline in the relative risk of stroke associated with diabetes from 3.1 in 1990 to 1.5 in 2010.²

Diabetes in Patients With Cerebrovascular Disease

Approximately 30% of patients with acute stroke have T1D or T2D diabetes including some who are previously undiagnosed.^{21,27,28} Epidemiologic studies typically do not distinguish between diabetes types, but 90% to 95% of cases will be T2D. Among those who do not have diabetes, 40% to 50% will have pre-diabetes as defined by a hemoglobin A1c (A1c) of 5.7% to 6.4% or by an oral glucose tolerance test^{27,28}; these persons are at high risk for developing diabetes in the subsequent 5 years compared with persons who have a normal A1c of <5.7% (10% risk compared with about 3%).²⁹ Undiagnosed diabetes and high risk for future diabetes is why professional organizations recommend regular screening after stroke.^{9,30}

With a few exceptions, epidemiologic studies show that patients with diabetes at the time of their stroke event have a higher risk for recurrent stroke, coronary heart disease, or death compared with patients without diabetes. According to a recent meta-analysis, patients with diabetes have a 50% increased risk for recurrence compared with a patient without diabetes (pooled hazard ratio [HR], 1.50 [95% CI, 1.36–1.65]).³¹ As for incident stroke in the general population, diabetes is likely one driver of disparities in stroke prognosis.³²

Pathophysiology

T1D leads to vascular disease primarily through hyperglycemia. The pathways from T2D to vascular disease are more complex and include the combined effects of conditions that precede or accompany T2D and hyperglycemia itself (Figure 1). The chief precursors to T2D are obesity and insulin resistance, the former being a component of the metabolic syndrome (ie, abdominal obesity, elevated blood pressure, elevated triglycerides, reduced HDL-cholesterol, and elevated glucose).^{33–35} The precursors and components of the syndrome are interrelated. Obesity contributes to insulin resistance, hypertension, dyslipidemia, and inflammation. Insulin resistance contributes to hypertension, dyslipidemia, endothelial dysfunction (including decreased generation of nitric oxide), vascular smooth muscle cell proliferation, inflammation, and hypercoagulability.

When insulin resistance is accompanied by concurrent relative insulin deficiency due to pancreatic beta-cell dysfunction, the result is hyperglycemia, further compounding the propensity toward atherosclerosis. Hyperglycemia increases vascular risk by several pathways including those shown in Figure 1.

Clinical Presentation and Diagnosis

Most neurologists will never diagnose T1D, which usually presents with acute symptoms requiring urgent medical attention, but almost every neurologist will diagnose T2D. Most commonly, they will find new T2D during screening of hospitalized patients or upon review of routine hospital laboratory testing. Because undiagnosed T2D is common among people with vascular disease, The American Diabetes Association (ADA) and the American Heart Association recommend screening for all adults with stroke at admission and periodically thereafter. About 20% of patients will have unrecognized diabetes at admission.^{27,30} The optimal interval for screening is uncertain, but the ADA suggests every 3 years for most patients.

A small proportion of patients with T2D are discovered because of symptoms of hyperglycemia such as polyuria, polydipsia, blurry vision, and fatigue. Some of these symptomatic patients will go on to the hyperosmolar hyperglycemic state or, rarely, diabetic ketoacidosis,

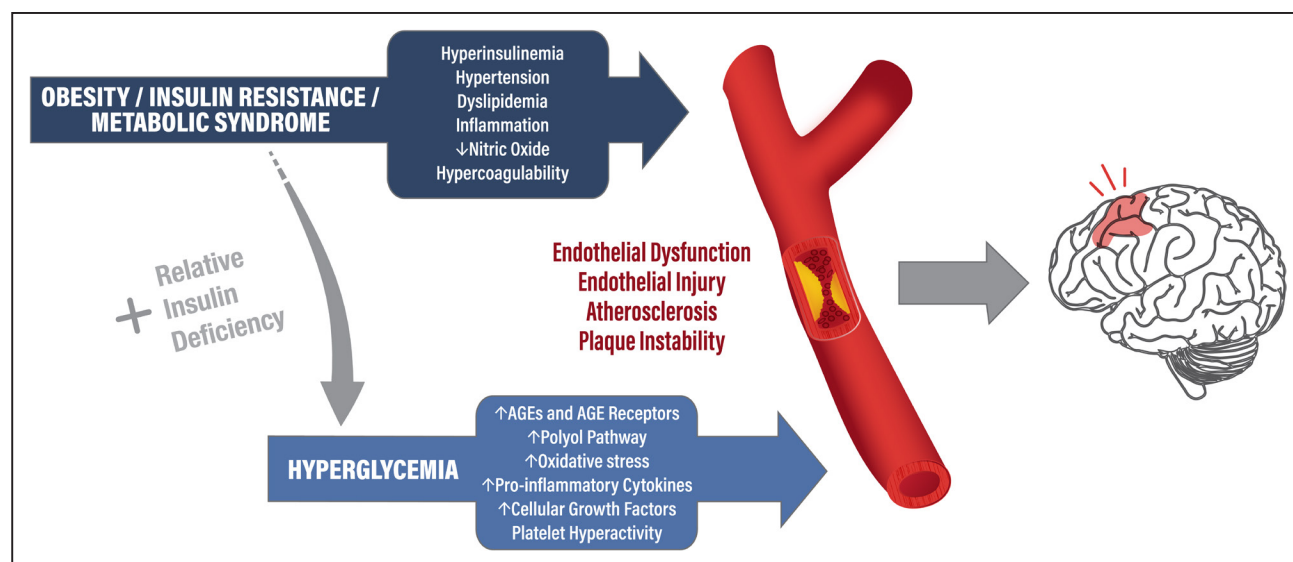


Figure 1. Mechanisms by which type 2 diabetes (T2D) may lead to vascular disease.

Precursors to diabetes include obesity and insulin resistance, the former being one component of the metabolic syndrome. The precursors and the syndrome contribute to endothelial injury, endothelial dysfunction, and atherosclerosis. When combined with relative insulin deficiency, insulin resistance causes the hyperglycemia that defines T2D. Hyperglycemia causes damage to blood vessels by several pathways shown. AGE indicates advanced glycation end products.

the latter of which may happen in the context of some other acute illness, such as sepsis. These symptoms should lead to prompt medical evaluation.

Management of Diabetes/Hyperglycemia in Patients With Acute Stroke

Admission hyperglycemia or stress hyperglycemia, especially in persons without diabetes, is associated with worse outcome for patients with acute stroke, ischemic or hemorrhagic.^{36–39} It reduces the likelihood of a favorable response to thrombolysis or mechanical thrombectomy and may increase the procedure-related risk of intracerebral hemorrhage.^{36,38,40–45}

It is no wonder that investigators reasoned that tight glucose control might improve outcome after stroke. When the theory was tested in a randomized trial, however, patients with acute ischemic stroke and T2D did not benefit from intensive glucose control (80–130 mg/dL) compared with conventional control (80–179 mg/dL) for the primary outcome of functional status at 90 days.⁴⁶ Tighter control was associated with increased risk for symptomatic hypoglycemia.

The American Heart Association/American Stroke Association and the ADA recommend serum glucose concentrations in the range of 140 to 180 mg/dL (7.8–10 mmol/L) during the first 24 hours after admission and until discharge for most patients with acute stroke.^{47,48} This applies to patients with either T1D or T2D. Most commonly, this is achieved with insulin. Oral medications for patients with T2D, particularly metformin and sulfonylureas, are typically held until a day or so prior

to discharge when patients are transitioned back to their home regimens.

Long-Term Management After Acute Stroke: Focus on T2D

Organization of Care

The goals of care for patients with T1D and T2D are the same: to prevent complications, preserve a normal quality of life, and avoid treatment-related adverse effects. Achieving these goals is often difficult and requires coordinated, multimodality care from well-prepared teams. Where should patients with T2D receive this care? For most patients, the answer is their primary care clinician. There are not enough endocrinologists for all patients with T2D, and primary care physicians and advanced care providers such as nurse practitioners and physician assistants are capable of treating most patients to goal.

More important than where patients are treated is how they are treated. Patients with T2D benefit from coordinated, comprehensive, continuous care to include health education, nutrition counseling, glucose control, early detection of complications, cardiovascular risk reduction and treatment of multimorbidity (eg, obesity, hypertension, dyslipidemia, chronic kidney disease). This sort of care falls within the bounds of the chronic disease management model.⁴⁹ When applied to diabetes, this model entails a patient-centered approach with shared decision making on all aspects of care, personal goal setting, self-management education and support, monitoring and feedback (eg, for glucose, blood pressure, weight), clinical informatics, and team based-care that is pro-active in

responding to care needs and early signs of treatment failure.^{49–51} Chronic disease management and multimorbidity are core features of effective primary care, making this specialty the appropriate home for most patients with T2D. For selected patients, particularly those with advanced complications, recurrent hypoglycemia, or needing complex insulin regimens, co-management with an endocrinologist can be very helpful.

Targets for Glycemic Control

The DCCT (Diabetes Control and Complications Trial) was the first study to test the effectiveness of intensive glucose control. DCCT randomized 1441 patients with T1D to intensive or conventional glucose control with insulin.³ Over a mean of 6.5 years, the A1c achieved in these 2 groups was $\approx 7\%$ versus 9% , respectively. While there were notable reductions with more intensive treatment in rates of microvascular complications such as retinopathy, nephropathy, and neuropathy, no statistically significant reduction in macrovascular events (including stroke) was observed in this relatively young population. In long-term follow-up (30 years) of the cohort beyond the randomized treatment phase, however, the incidence of major adverse cardiovascular events (MACE=myocardial infarction, stroke, or cardiovascular death) was non-significantly lower in the intensive treatment group (RR, 0.68; $P=0.07$).⁵² There was no signal of benefit for stroke prevention.

In T2D, the impact of glycemic control on macrovascular disease has also been disappointing. The UKPDS (United Kingdom Prospective Diabetes Study) involved 4209 newly diagnosed patients with T2D.⁴ The more intensive arm incorporated treatment with sulfonylureas or insulin. During the trial, the mean A1c was 7.0% versus 7.9% , respectively. Similar to the DCCT, intensive therapy was associated with fewer microvascular outcomes, but no significant difference in cardiovascular events. With the accumulation of more events in a long-term follow-up study, the incidence of myocardial infarction was significantly less in the intensive arm, compared with the conventional arm (HR, 0.85; $P=0.01$). For stroke, however, any difference was not statistically significant (HR, 0.91; $P=0.39$).

In 2008 to 2009, 3 additional cardiovascular outcomes trials in patients with T2D published their findings for the effect of stringent diabetes control (A1c target of $<6\%$ to 6.5%), compared with standard control, for prevention of cardiovascular outcomes.^{53–55} None demonstrated a significant benefit for more stringent glucose control on MACE or stroke alone.

As a result of the DCCT, UKPDS, and the more recent cardiovascular outcome trials, we can conclude that targeting tighter glucose control does not clearly reduce the incidence of macrovascular events, including stroke, in patients with T1D or T2D. These studies underlie professional recommendations for a A1c $<7\%$ for most

adults to reduce risk for microvascular disease; a lower goal might be appropriate for a younger, healthier persons and a higher goal might be best for an older person, someone with significant comorbidity, or someone with high risk for hypoglycemia.^{56,57}

Nonpharmaceutical Interventions for Glucose Control

Diet, physical activity, and weight management can improve the underlying causes of T2D and have a central role in glucose control.⁵⁸ Weight loss is particularly important because it can result in disease remission or reduce the number/dose of glucose-lowering medications.¹³ No lifestyle intervention (ie, diet, physical activity, weight management) has been shown in a randomized clinical trial to prevent macrovascular disease in T2D, but there is enough experimental data on glucose control to recommend them as fundamental, safe elements of disease management; plus, they have benefits beyond glycemia itself.

To optimize diet, activity, and weight, patients with T2D should have professional encouragement and assistance, including regular formal diabetes self-management education and support from a certified instructor. The curriculum for diabetes self-management education and support covers, among other topics, healthy eating, physical activity, medication usage, and glucose monitoring; diabetes self-management education and support has been shown to improve glucose control.⁵¹ Beyond diabetes self-management education and support, physician advice for lifestyle improvement can have a moderate effect on behavior and should always be provided. More formal counseling for a healthy diet and physical activity can also have a moderate benefit but availability will depend on local resources. All persons with obesity should be offered formal management in the form of a program or practice that can provide counseling, medications, and advice on bariatric (sometimes referred to as “metabolic”) surgery.⁵⁹ Weight counseling on its own can have a modest effect on weight. Examples of counseling programs for weight loss include the Diabetes Prevention Program, which is often available from local organizations.⁶⁰ When counseling for dietary modification alone is not adequate, new medications are safe and effective for achieving significant weight loss and improved glucose control.⁶¹ When counseling and medications fail, bariatric surgery is highly effective for glucose control and (sometimes) disease remission for appropriate patients.⁶²

Pharmacotherapy

At least one agent in each of 13 pharmaceutical classes are approved by the US Food and Drug Administration for glucose control in adults with T2D; agents from 7 classes are in widespread use (Table).^{51,57,58,63}

The ADA and other organizations have published recommendations for sequencing and combining these medications.^{51,56,58} These recommendations are largely

Table. Selected Characteristics of Medications From 7 Pharmaceutical Classes in Widespread Use for T2D*

Class	Specific agents	Main mechanisms of action	Advantages		Disadvantages/adverse effects	Basic dosing/monitoring
			General	For prevention of CVD/stroke		
Oral antihyperglycemic medications						
Sulfonylureas	Glyburide Glipizide Glimepiride	↑ Insulin secretion	Extensive experience Inexpensive	No proven benefit	Hypoglycemia risk	Take 30 min before meals; start 3 mg once daily Avoid in elderly and some disabled persons due to hypoglycemic risk A1c every 3–6 mo
Biguanides	Metformin	↓ Hepatic glucose production	Extensive experience No/rare hypoglycemia† Inexpensive	Signal of possible benefit for stroke prevention from UKPDS	Gastrointestinal symptoms Vitamin B12 deficiency Lactic acidosis (rare)	Start with 500 mg daily, increase gradually based in A1c goal. Max 2000 mg daily Avoid in advanced renal disease Monitor plasma B12 Hold before iodinated contrast procedures if eGFR<60 Avoid in advanced liver disease A1c every 3–6 mo
Thiazolidinediones	Pioglitazone	↑Insulin sensitivity	No/rare hypoglycemia† Inexpensive	Proven benefit in patients with stroke and insulin resistance and possibly in high-risk patients with diabetes	Weight gain Dependent edema Exacerbation of heart failure Bone fracture Possible ↑ risk bladder cancer Do not use in patients with heart failure, osteoporosis, or bladder cancer	Start 15 mg, max 45 mg Monitor weight, edema A1c every 3–6 mo
DPP-4 inhibitors	Alogliptin Linagliptin Saxagliptin Sitagliptin	↑ GLP-1 and GIP ↑ Insulin secretion ↓ Glucagon secretion	No/rare hypoglycemia† Well tolerated	Some agents proven neutral on CVD outcomes	Lower efficacy in lowering glucose than other classes ? Angioedema ? Pancreatitis High cost	Dose varies by agent (some require adjustments in CKD) Do not combine with GLP-1 RAs A1c every 3–6 mo
GLP-1 receptor agonists	Semaglutide	↑ GLP-1 and GIP ↑ Insulin secretion ↓ Glucagon secretion Slow gastric emptying	No/rare hypoglycemia† Weight loss	Members of this class have been associated with reductions in MACE, with a seemingly larger effect on prevention of stroke	Gastrointestinal symptoms Pancreatitis and cholelithiasis have been reported High cost ? ↑ Risk retinopathy complications in patients with preexisting retinopathy and high A1c	Take 30 min before meals; start 3 mg once daily Do not use with personal/family history of medullary thyroid carcinoma or MEN2 A1c every 3–6 mo
SGLT2 inhibitors	Canagliflozin Dapagliflozin Empagliflozin Ertugliflozin	↓ Renal glucose reabsorption	No/rare hypoglycemia† Weight loss ↓ Blood pressure	Members of this class have been associated with reductions in MACE, HF hospitalizations, and CKD progression	↓ Volume Genital infections ? Amputation risk (canagliflozin) ? Fracture risk (canagliflozin) ? UTI Polyuria High cost Slight ↑ risk DKA	Dose varies by agent Adjust dose or avoid for CKD Watch for volume depletion A1c every 3–6 mo
Injectable antihyperglycemic medications						
GLP-1 receptor agonists	Dulaglutide Exenatide Liraglutide Lixisenatide Semaglutide Tirzepatide‡	↑ GLP-1 activity ↑ Insulin secretion ↓ Glucagon secretion Slows gastric emptying	No/rare hypoglycemia† Weight loss	Members of this class have been associated with reductions in MACE, with a seemingly larger effect on prevention of stroke	Gastrointestinal symptoms Pancreatitis and cholelithiasis have been reported High cost	Initial dose varies by agent, start low Do not use with personal/family history of medullary thyroid carcinoma or MEN2 A1c every 3–6 mo

(Continued)

Table. Continued

Class	Specific agents	Main mechanisms of action	Advantages		Disadvantages/adverse effects	Basic dosing/monitoring
			General	For prevention of CVD/stroke		
Insulin	Many	↑ Peripheral glucose uptake ↓ Hepatic glucose production	Every patient will respond No ceiling effect on reducing glucose	No proven benefit Some agents shown to be neutral	Hypoglycemic risk Weight gain May require frequent injections Some agents high cost Requires careful patient education	Dosing varies Avoid in persons at high-risk of hypoglycemia A1c every 3–6 mo

CHF indicates congestive heart failure; CKD, chronic kidney disease; CV, cardiovascular; CVD, cardiovascular disease; GLP-1, glucagon-like peptide 1; MACE, major adverse cardiovascular event (ie, nonfatal stroke, nonfatal myocardial infarction, cardiovascular death); MEN2, multiple endocrine neoplasia syndrome type 2; MI, myocardial infarction; T2D, type 2 diabetes; and UKPDS, United Kingdom Prospective Diabetes Study.

*Table based partly on consensus statements, guidelines, and reviews.^{51,57,58,63}
†Hypoglycemia unlikely with monotherapy, but the drug can contribute to hypoglycemia when used with other agents.
‡Tirzepatide is a dual receptor agonist for glucose-dependent insulinotropic polypeptide and glucagon-like peptide-1.

consistent in emphasizing a patient-centered approach in which glucose goals and treatment selection accounts for each patient's preference, values, and co-morbid conditions, especially renal disease and heart failure. For patients with stroke, the ADA and the European Association for the Study of Diabetes recommend initial therapy for most patients that includes a GLP-1 receptor agonist or SGLT2 inhibitor and consideration of pioglitazone. Other groups prefer GLP-1 receptor agonists first in patients with atherosclerotic cardiovascular disease or indicators of high risk, including patients with ischemic stroke. SGLT2 inhibitors are preferred for patients with heart failure or chronic kidney disease, especially when the latter is accompanied by albuminuria. GLP-1 receptor agonists and SGLT2 inhibitors can be combined to achieve A1c goals. Pioglitazone can be added for A1C control in patients with atherosclerotic cardiovascular disease who do not have heart failure. Many organizations, including the ADA, also recommend metformin for initial drug therapy. Importantly, in contrast to prior guidelines, the addition of GLP-1 receptor agonists or SGLT2 inhibitors should be made based on comorbidities, not solely the need for additional glucose lowering beyond metformin monotherapy.

Guidelines now promote GLP-1 receptor agonists, SGLT2 inhibitors, and pioglitazone for patients with stroke or cardiovascular disease because they are the only 3 agents shown to both reduce glucose and risk for MACE in patients with these conditions. Because of their importance for patient with stroke and because these data are relatively recent, we provide more details here on their clinical use.

GLP-1 Receptor Agonists

GLP-1 receptor agonists increase plasma levels of GLP-1 which increases glucose-mediated insulin secretion, suppresses glucagon secretion, slows gastric emptying, and increases satiety. The GLP-1 receptor agonists are available for oral (semaglutide) or injection (all agents) administration. They are associated with

dyspepsia and nausea, but these symptoms rarely lead to discontinuation. They should not be used or avoided in patients with pancreatitis, medullary thyroid cancer, gastroparesis, or increased risk for nausea. They should not be combined with DPP-4 inhibitors.

Three agents from this class (ie, liraglutide, dulaglutide, semaglutide) have been shown to prevent MACE in patients with T2D who also have cardiovascular disease or high risk for cardiovascular disease.⁶⁴ In a large cardiovascular outcome trial, dulaglutide reduced risk for stroke by 24% ($P=0.01$).⁶⁵ In the largest meta-analysis of GLP-1 receptor agonists, incorporating data from 8 T2D cardiovascular outcome trials, the relative risk reduction estimates were 14% for MACE, 13% for cardiovascular mortality, and 17% for fatal/nonfatal stroke (all $P<0.01$).⁶⁴

A separate meta-analysis focused on just stroke outcomes. In 8 trials, GLP-1 receptor agonist therapy was associated with reduced risk for any stroke (odds ratio [OR], 0.84 [95% CI, 0.75–0.93]) and non-fatal strokes (OR, 0.84 [95% CI, 0.76–0.94]), but not fatal strokes (OR, 0.82 [95% CI, 0.62–1.09]).⁶⁶ A secondary analysis was conducted for risk of non-fatal stroke in those with a prior history of myocardial infarction or stroke; the analysis was limited to 2 trials and the effect was not significant (OR, 0.87 [95% CI, 0.67–1.15]).

SGLT2 Inhibitors

The SGLT2 inhibitors block glucose reabsorption by the nephron and therefore cause glycosuria. They are dispensed in oral form and are safe but can increase risk for mycotic genital infections. In large cardiovascular outcome trials, they modestly reduced risk for MACE and cardiovascular mortality and more impressively reduce risk for heart failure and progression of chronic kidney disease. Generally, the cardio-renal benefits from this drug category appear to be mediated by hemodynamic factors rather than reductions in atherosclerosis. According to a recent meta-analysis, SGLT2 inhibitors significantly reduce risk for MACE (10%), cardiovascular

mortality (15%), HF hospitalization (32%), and chronic kidney disease progression (38%; all $P < 0.05$).⁶⁷ In contrast to the GLP-1 receptor agonists, they have no apparent effect on stroke prevention (HR, 0.96; $P = 0.29$). It should be noted, however, that patients with a history of stroke appeared to derive the same benefit as other patients in terms of MACE, HF hospitalizations, and chronic kidney disease progression.⁶⁸

Pioglitazone

Pioglitazone is an inexpensive oral medication that lowers glucose by improving insulin resistance. As monotherapy, it does not cause hypoglycemia, but adverse events have limited its clinical use. Weight gain and dependent edema respond to dose reduction. Heart failure due to fluid retention can occur but is rare without preexisting heart disease. Bone fracture risk is increased about 50% after 5 years of therapy. Pioglitazone should not be used in patients with heart failure, osteoporosis, or bladder cancer.

Pioglitazone was the first glucose-lowering agent to show promise for prevention of macrovascular disease, including stroke. In the PROactive (PROspective pioglitazone Clinical Trial in macroVascular Events) study of patients with T2D and macrovascular disease, pioglitazone was associated with a 16% relative risk reduction for all-cause mortality, myocardial infarction or stroke compared with placebo (HR, 0.84 [95% CI, 0.72–0.98]; $P = 0.027$).⁶⁹ Pioglitazone was later tested in the IRIS trial (Insulin Resistance Intervention after Stroke), the only dedicated stroke secondary prevention study to involve a glucose-lowering drug.⁷⁰ IRIS enrolled 3876 patients with a recent stroke or transient ischemic attack who had insulin resistance but no diabetes. After 4.8 years, the risk for myocardial infarction or stroke was reduced by 24% among patients assigned to pioglitazone (HR, 0.76 [95% CI, 0.62–0.93]). The risk for stroke alone was reduced by 25% (HR, 0.75 [95% CI, 0.60–0.94]), driven by a 28% relative risk reduction in ischemic stroke (HR, 0.72 [95% CI, 0.57–0.91]).⁷¹

Other Therapies for Cardiovascular Risk Reduction

Other than lifestyle and glucose-lowering therapy, patients with diabetes after stroke should receive the same secondary prevention therapy as patients without diabetes.³⁰ LDL cholesterol reduction with a statin drug, blood pressure reduction, antithrombotic therapy, anticoagulation for atrial fibrillation, and carotid endarterectomy are all important. The evidence for these therapies in diabetes is from stroke secondary prevention trials which enrolled patients with and without diabetes. Not all trials reported effects in subgroups defined by diabetes; among those that did only the Canadian aspirin trial reported a diabetes-treatment group interaction. In that study, the benefit of aspirin was only seen in men without diabetes.⁷² The advice regarding cholesterol

reduction and blood pressure control is strengthened by primary prevention trials in patients with diabetes^{5,7} and a trial of intensive multifactorial management in high-risk patients.⁸

Long-Term Management After Acute Stroke: Focus on T1D

Patients with T1D are almost always managed by an endocrinologist because of their dependence on continuous subcutaneous insulin or multiple daily injections. Diet, physical activity, and weight management are important for hour-to-hour and long-term glucose control, but these non-pharmaceutical interventions do not significantly change the underlying causes of T1D.

Insulin is the only glucose-lowering therapy approved for patients with T1D. It is the oldest and most effective agent in lowering glucose but carries a risk for hypoglycemia. Based on evidence discussed above from the DCCT and UKPDS trials and a more recent cardiovascular outcomes study,⁷³ insulin does not seem to prevent MACE, stroke, or other cardiovascular outcomes. Other therapies for cardiovascular risk reduction (see above) are the same for patients with T1D and T2D.

An Approach to Diabetes Management in Neurology Practice

Neurologists can improve diabetes management at every patient contact, but 3 opportunities may be most important:

1. In the Hospital. The focus of diabetes management in the hospital is glycemic control, usually with insulin. Rarely is the hospital an appropriate occasion to make substantial changes in a patient's home therapy regimen, but it can be an opportunity to educate patients and families on the goals of diabetes care, the options for achieving goals, and the value of optimal management. Most commonly, patients resume their home medication regimen a day or 2 before discharge.
2. The Transition from Hospital to Home or Rehabilitation Facility. This transition is potentially risky, but clinicians can mitigate the risks by planning. Most important is to assure that patients have resources to achieve their diabetes goals including access to a treating clinician. They should leave with an appointment with that person; for T1D this is typically an endocrinologist but for T2D it is typically the primary care clinician.⁷⁴ Hospital personnel should communicate with this clinician to review the hospital course and discharge plans. Some stroke programs call patients at home after discharge to prevent lapses in care.
3. In the Office. The most important component of diabetes-related care in neurology office-based

practice is collaboration with primary care (for most patients with T2D), endocrinology (for most patients with T1D and some with T2D, particularly if on complex insulin regimens), and community-based services. This collaboration is operationalized in the 3-question strategy we recommend for patients with T2D (Figure 2).^{58,75,76} The strategy is based partly on research showing that collaboration between neurology professionals and primary care teams can improve risk factor control.⁷⁴ It is also based on models of chronic disease management⁴⁹ and recommendations from several professional groups, including the ADA.^{30,50,56,58,77} The strategy presumes that neurology clinicians will advocate for optimal management, but not necessarily prescribe diabetes medications after hospital discharge.

Question 1: “Who is managing the patient’s diabetes?” An answer indicating no treating clinician or no recent visit represents a lapse in care that would need to be addressed.

Question 2: “Is A1c at target?” An A1c that is missing or above target can indicate non-adherence or lack of access to care. Explain the benefits of monitoring if not done recently. If the A1 is above goal, explore reasons (eg, medication non-adherence, missed medical appointments, depression) and arrange visit with diabetes clinician.

Question 3: “Is diabetes strategy optimized?” There are no cardiovascular outcome trials in patients with stroke and diabetes, but the optimal regimen, according to ADA guidelines, should include a GLP-1 receptor agonist or an SGLT2 inhibitor on a background of lifestyle optimization and, usually,

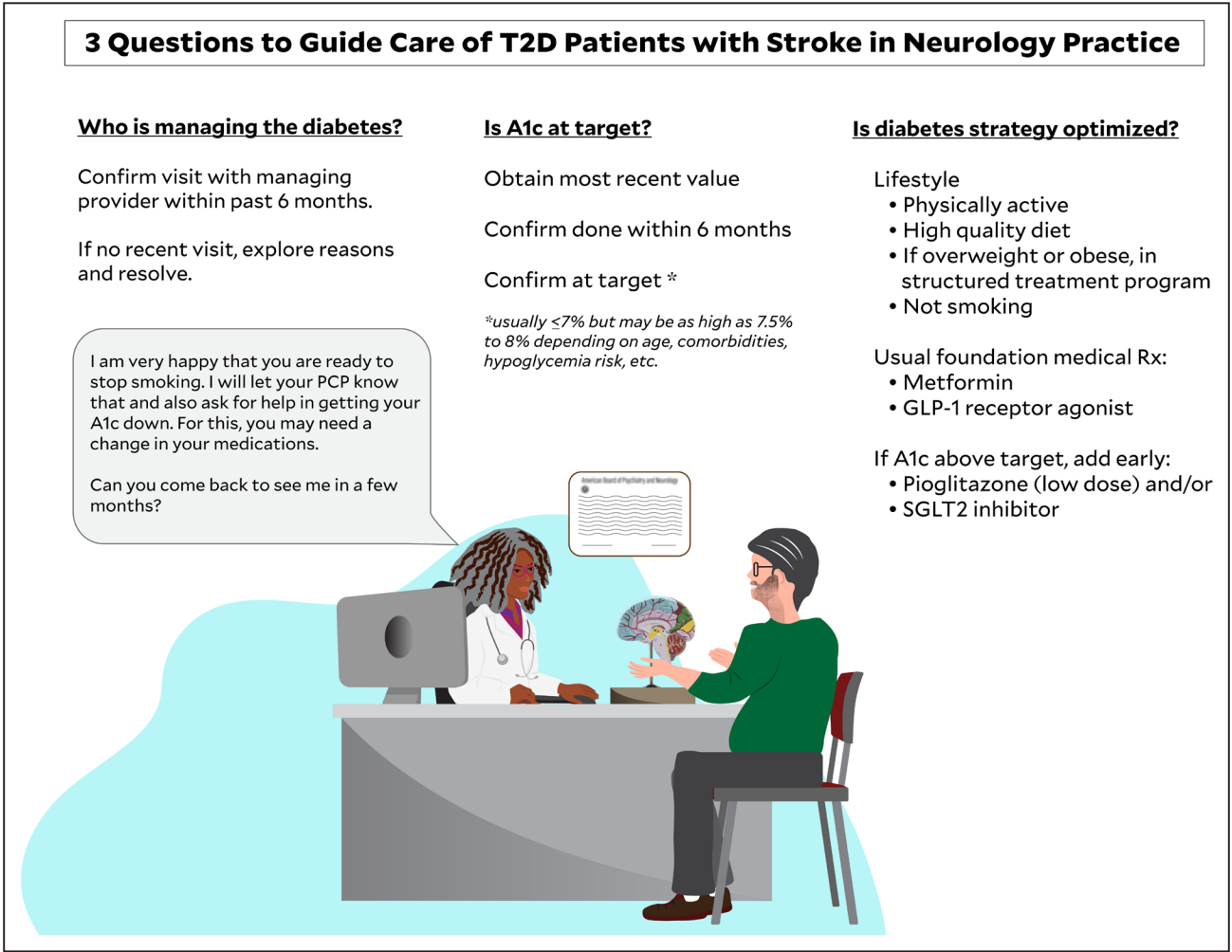


Figure 2. Proposed strategy for addressing type 2 diabetes (T2D) in neurology practice. Most neurology clinicians do not regularly prescribe diabetes medications after hospital discharge, but all can promote diabetes care by asking the 3 questions shown. Actions steps to consider, based on answers, are described in the text. Unlike the ADA guideline,⁵⁸ this algorithm suggests prioritizing GLP-1 (glucagon-like peptide 1) receptor agonists as part of foundational therapy rather than giving equal considering to SGLT2 (sodium glucose co-transporter 2) inhibitors because the evidence for stroke reduction is only present with the former. Note that SGLT2 inhibitor therapy is recommended in T2D patients with heart failure⁷⁵ or chronic kidney disease (so long as estimated glomerular filtration rate ≥ 25 –30 mL/min/1.73 m²).⁷⁶ Pioglitazone is listed because of its benefit in reducing cardiovascular events, including stroke, and because it is the only glucose-lowering agent tested in a stroke trial (IRIS).

metformin. Neurologists should monitor weight and inquire about lifestyle including physical activity, smoking, and diet quality. When overweight or lifestyle problems are identified, it is appropriate to call these to the attention of patients as opportunities to improve brain health and refer to the primary care clinician for assistance. The ADA guidelines do not represent that GLP-1 receptor agonists and SGLT2 inhibitors have differing effects on stroke incidence but only the former is associated with stroke prevention.⁷⁸ We believe that most patients with diabetes and stroke should take a GLP-1 receptor agonist if they do not enter remission with lifestyle therapy and weight loss. If the A1c goal is not achieved on foundational therapy, intensification is warranted. Among options, SGLT2 inhibitors and pioglitazone reduce risk for vascular events in patients with atherosclerotic cardiovascular disease, including stroke, but only pioglitazone reduces stroke specifically.

During hospitalization, at transition from the hospital, and in the office, it is important to identify personal barriers to optimal diabetes care. These can usually be uncovered by compassionate inquiry. Among barriers, the social determinants of health (ie, socioeconomic status, neighborhood and physical environment, food environment, health care, and social context) are important and warrant discovery.⁷⁹ Once adverse social determinants are discovered, they can be mitigated by emerging practice-based interventions. Examples include DSME tailored to language and culture, case management by nurses, pharmacist-led treatment, and social support enhancement. If these are not available in a neurology practice, that is another reason for collaboration with primary care or endocrinology.

FINAL COMMENT

Proper management of diabetes can improve the health of patients with stroke and reduce their risk for further brain injury. Neurology clinicians can help by knowing current diabetes recommendations, monitoring the care of their patients, and advocating or directly adjusting therapy when indicated.

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Disclosures

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